

Synoptic CB: Soil Fluxes

2024 Samples

2026-07-27

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#Read in and collate all the files in the Processed Data folder

##pull TEROS data for the zone during the time of the fluxes and put it into another column so that we can use that too.

0.1 Match up the TEROS soil data and add it to the flux dataframe

##Remove bad fluxes and calculate how many there were

Number flux measurements removed: 17

Number negative CO2 fluxes removed: 5

Number fluxes with ebullition removed: 9

Number messy fluxes removed: 3

Percentage flux measurements removed: 2.26%

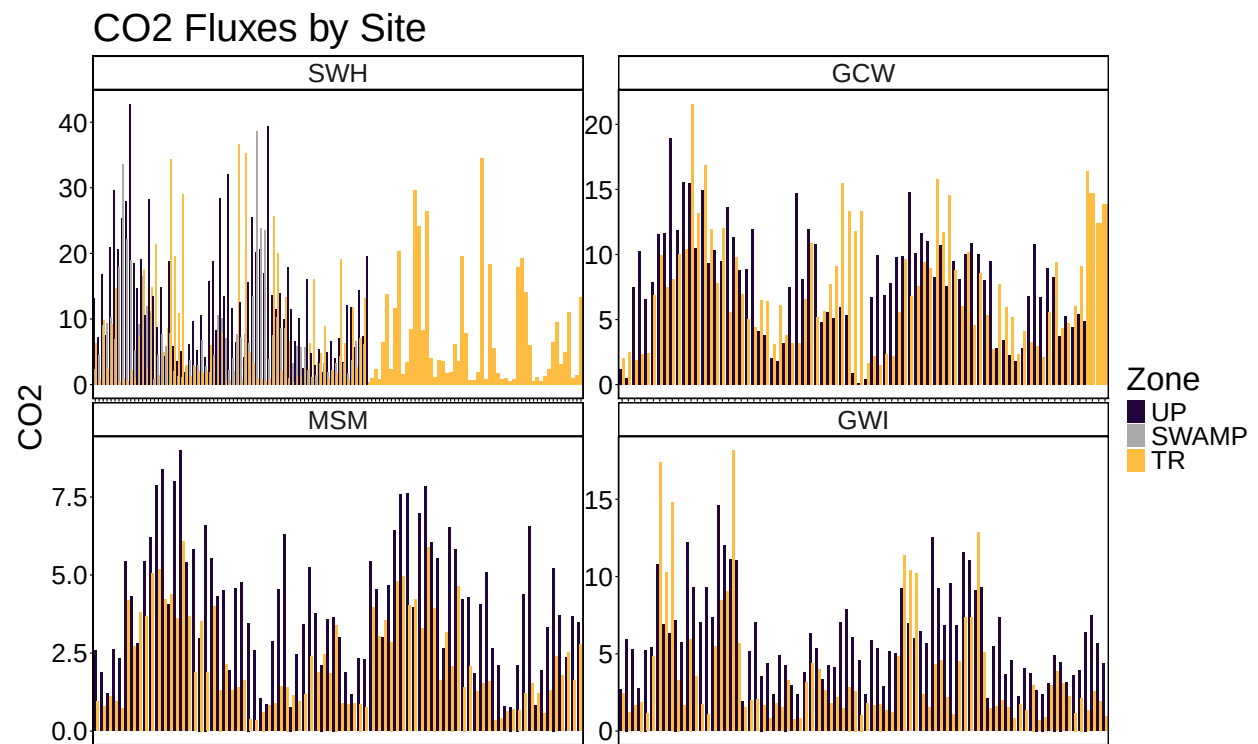
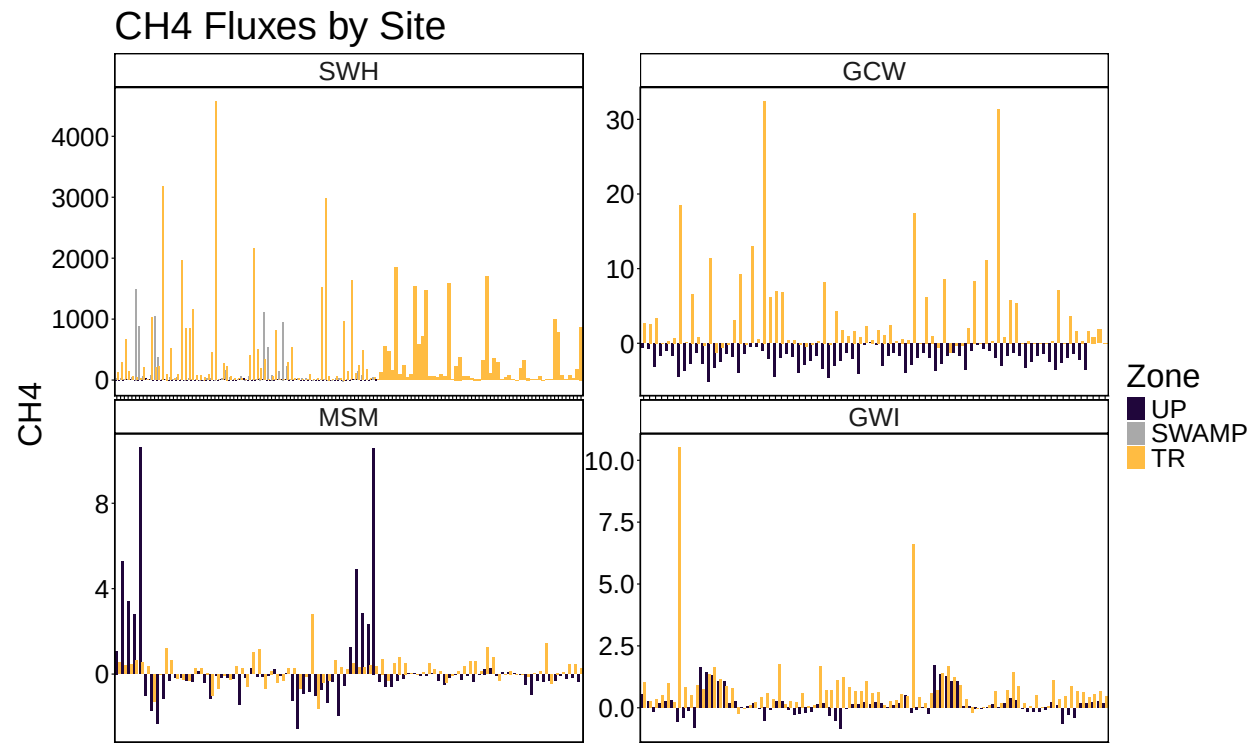
Percentage negative CO2 fluxes removed: 0.66%

Percentage fluxes with ebullition removed: 1.2%

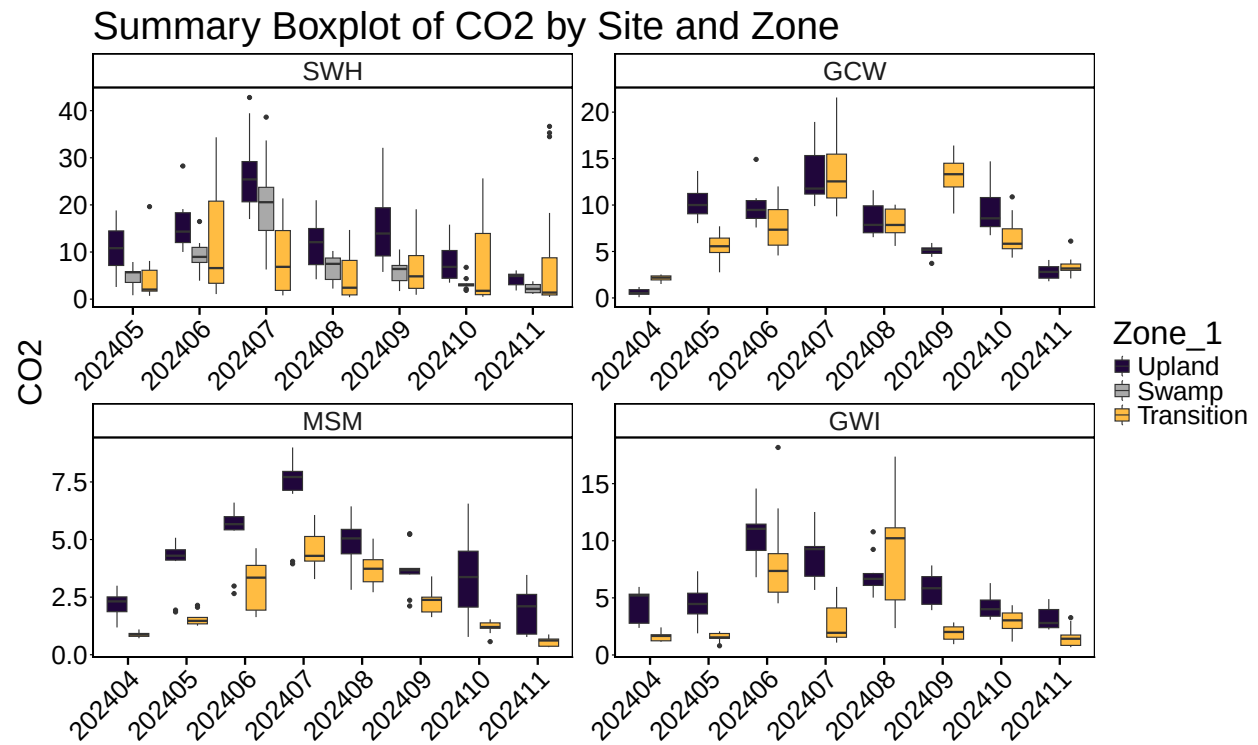
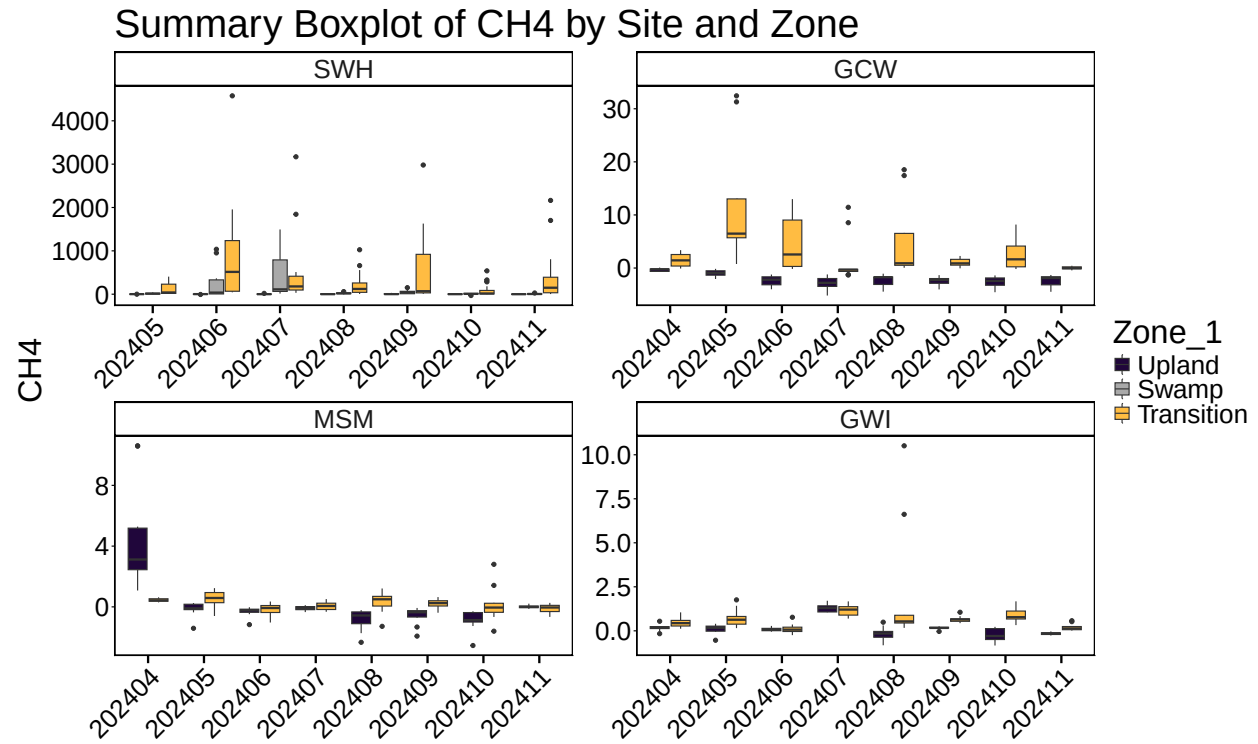
Percentage messy fluxes removed: 0.4%

0.2 Write out files

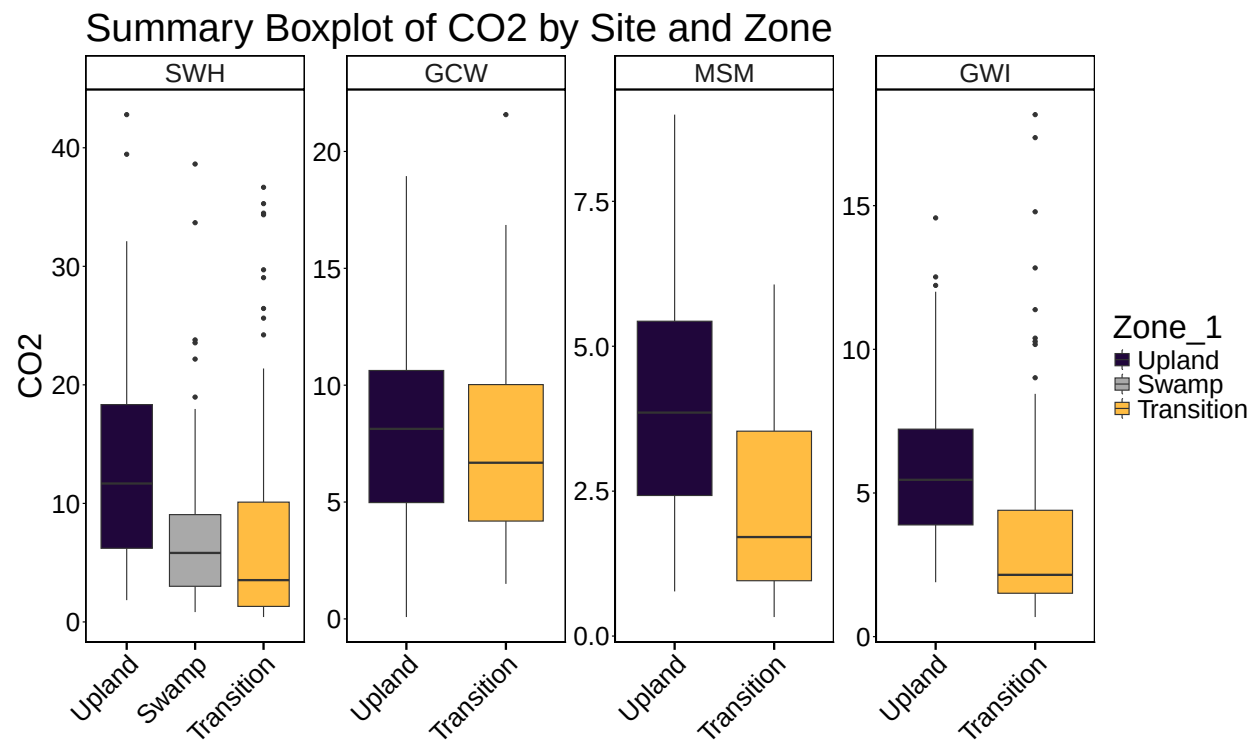
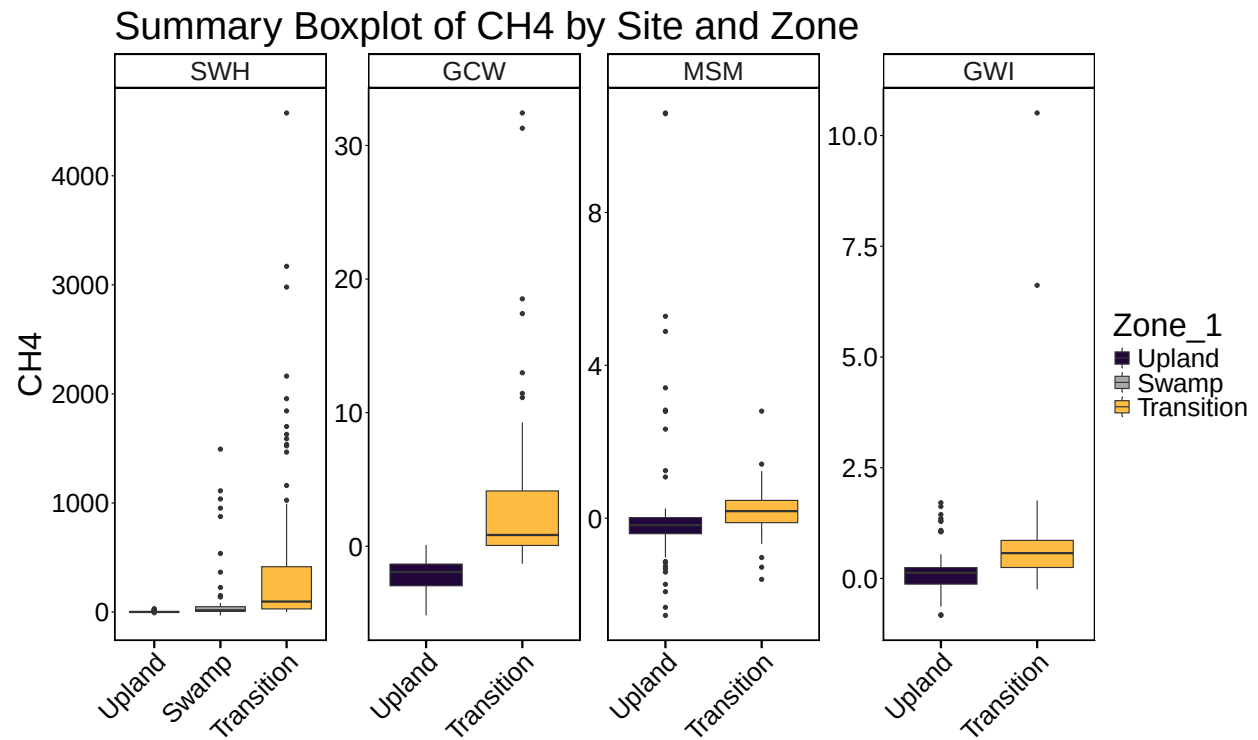
0.3 All Data by Year, Site and Zone



0.4 Boxplot summary by Date



0.5 Boxplot summary by Site



##Calculate summary metrics

0.6 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Upland	-6.8413713	30.5529007	2.6298108	1.8376000	42.796305	13.315112
SWH	Swamp	-30.8788013	1493.5428344	118.0735925	0.8299256	38.629648	7.589840
SWH	Transition	-0.4155641	4575.9598545	393.4634503	0.4119879	36.665947	7.444425
GCW	Upland	-5.1805432	0.0863896	-2.1612615	0.0795986	18.945223	7.919248
GCW	Transition	-1.3132659	32.4350379	3.3663421	1.4992124	21.575400	7.525248
MSM	Upland	-2.5440889	10.6033414	0.1982541	0.7701261	8.995517	4.046136
MSM	Transition	-1.6014901	2.8012684	0.1736025	0.3279700	6.065279	2.234828
GWI	Upland	-0.8341486	1.7090103	0.1568745	1.8942486	14.573228	6.022340
GWI	Transition	-0.2445658	10.5092327	0.7952364	0.6825014	18.167895	3.706339